

Low Acrylonitrile NBR

POLYMERIZATION/SYNTHESIS PATENT RESEARCH REPORT

1. ABSTRACT

This report provides a technical summary of the research scope, definition, and patent landscape of Low Acrylonitrile NBR (Nitrile Butadiene Rubber). The target of this research is to synthesize and analyze the polymerization approaches, formulation, and process trends for Low Acrylonitrile NBR. The selected patent landscape includes US and EP patents, which provide valuable insights into the synthesis and production of Low Acrylonitrile NBR. The major polymerization approaches include emulsion polymerization, and the major formulation/process trends include the use of specific monomer ratios, emulsifiers, initiators, and chain-transfer agents. However, the extracted patent evidence lacks specific information on the acrylonitrile content and butadiene content, which are crucial for the production of Low Acrylonitrile NBR.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

US3822242

Patent Details: - Patent Number: US3822242 - Patent Title: Manufacturing process for alternating copolymer of butadiene and acrylonitrile ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 1974-07-02 - Polymer Type: Nitrile Butadiene Rubber - Relevance to Low Acrylonitrile NBR: The patent provides a general procedure for the production of nitrile rubber, which can be used as a basis for the production of Low Acrylonitrile NBR.

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: 8.07 - Pressure: 16.0 - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed

patent evidence. - Conversion: 33.1 - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Yield: 22.9, Viscosity: 50.9

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it provides a general procedure for the production of nitrile rubber, which can be used as a basis for the production of Low Acrylonitrile NBR. However, the lack of specific information on the acrylonitrile content and butadiene content limits its technical usefulness.

EP3604397

Patent Details: - Patent Number: EP3604397 - Patent Title: PROCESS FOR THE PRODUCTION OF NITRILE RUBBER CONTAINING CARBOXYL GROUPS - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2020-11-18 - Polymer Type: Nitrile Butadiene Rubber - Relevance to Low Acrylonitrile NBR: The patent provides a process for the production of nitrile rubber containing carboxyl groups, which can be used as a basis for the production of Low Acrylonitrile NBR.

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it provides a process for the production of nitrile rubber containing carboxyl groups, which can be used as a basis for the production of Low Acrylonitrile NBR. However, the lack of specific information on the acrylonitrile content and butadiene content limits its technical usefulness.

EP2174723

Patent Details: - Patent Number: EP2174723 - Patent Title: Method for producing nitrile rubber metal laminate - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2014-03-12 - Polymer Type: Nitrile Butadiene Rubber - Relevance to Low Acrylonitrile NBR: The patent provides a method for producing nitrile rubber metal laminate, which can be used as a basis for

the production of Low Acrylonitrile NBR.

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: 20 - Emulsifier: Alkoxysilane - Emulsifier loading: 2.5 - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: 150 - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it provides a method for producing nitrile rubber metal laminate, which can be used as a basis for the production of Low Acrylonitrile NBR. However, the lack of specific information on the acrylonitrile content and butadiene content limits its technical usefulness.

US20100190018

Patent Details: - Patent Number: US20100190018 - Patent Title: Process for producing nitrile rubber-metal laminate - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2010-07-29 - Polymer Type: Nitrile Butadiene Rubber - Relevance to Low Acrylonitrile NBR: The patent provides a process for producing nitrile rubber-metal laminate, which can be used as a basis for the production of Low Acrylonitrile NBR.

Polymerization / Synthesis Method: - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Acrylonitrile content: Not disclosed in the analyzed patent evidence. - Butadiene content: Not disclosed in the analyzed patent evidence. - Exact monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: 20 - Emulsifier: Alkoxysilane - Emulsifier loading: 2.5 - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: 2.5 - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Technical relevance: This patent is relevant to Low Acrylonitrile NBR as it provides a process for producing nitrile rubber-metal laminate, which can be used as a basis for the production of Low Acrylonitrile NBR. However, the lack of specific information on the acrylonitrile content and butadiene content limits its technical usefulness.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

The selected patents provide limited information on the synthesis and production of Low Acrylonitrile NBR. The patents US3822242, EP3604397, EP2174723, and US20100190018 provide some information on the polymerization process, monomer system, and emulsifier loading, but lack specific information on the acrylonitrile content and butadiene content. * The polymerization temperature ranges from 2.5 to 150. * The emulsifier loading is 2.5 for EP2174723 and US20100190018. * The water amount is 20 for EP2174723 and US20100190018. * The conversion is 33.1 for US3822242. The lack of specific information on the acrylonitrile content and butadiene content limits the technical usefulness of these patents. However, the patents provide a basis for the production of Low Acrylonitrile NBR and can be used as a starting point for further research and development.

4. REFERENCES

- [US3822242](#) - Manufacturing process for alternating copolymer of butadiene and acrylonitrile ...
- [EP3604397](#) - PROCESS FOR THE PRODUCTION OF NITRILE RUBBER CONTAINING CARBOXYL GROUPS
- [EP2174723](#) - Method for producing nitrile rubber metal laminate
- [US20100190018](#) - Process for producing nitrile rubber-metal laminate